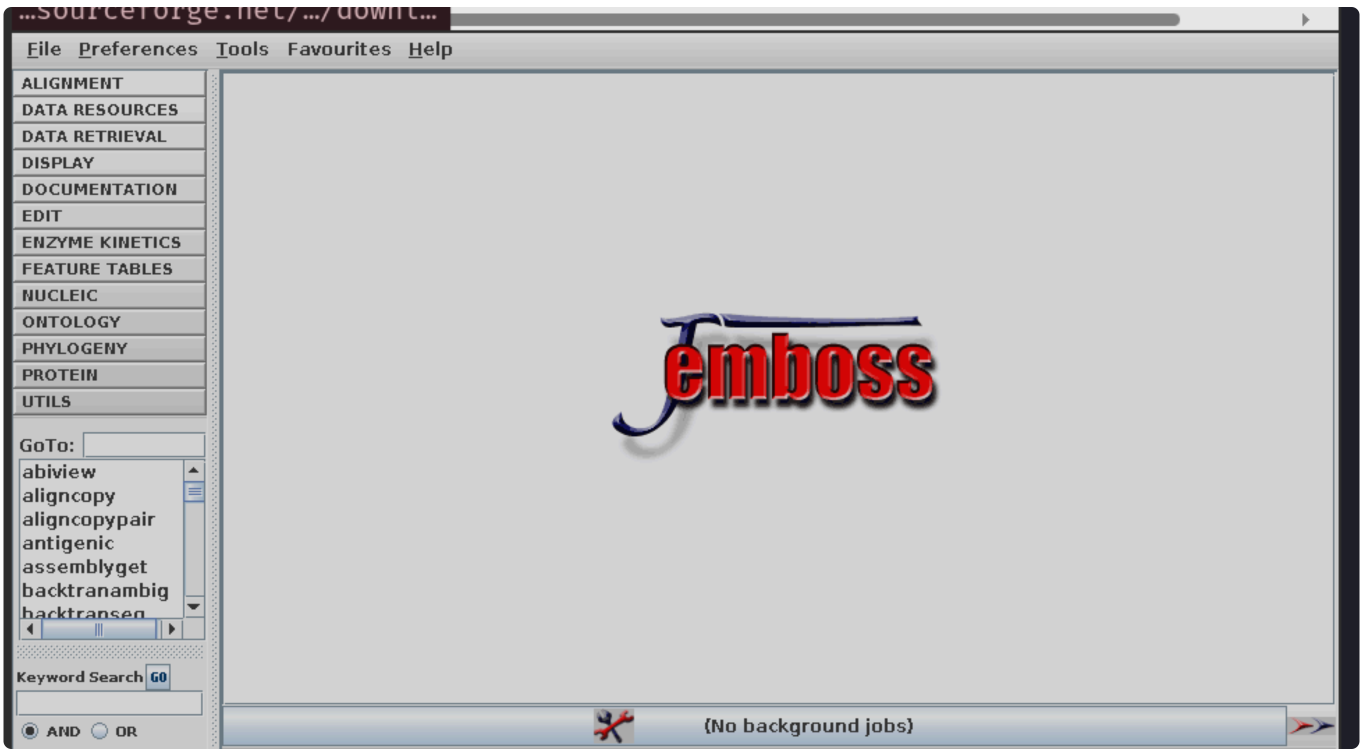
Solution -1

all csvs and python file are attached here
https://github.com/specapoorv/BT3040_BioInfo_Lab_assignments/tree/main/assignment_1

1. Download the EMBOSS package (http://emboss.sourceforge.net/download/) and copy to your Windows system. In case of **Linux** use the command: **sudo apt-get install jemboss**. For Mac users, use the online tool links given in the instructions doc.

solution :



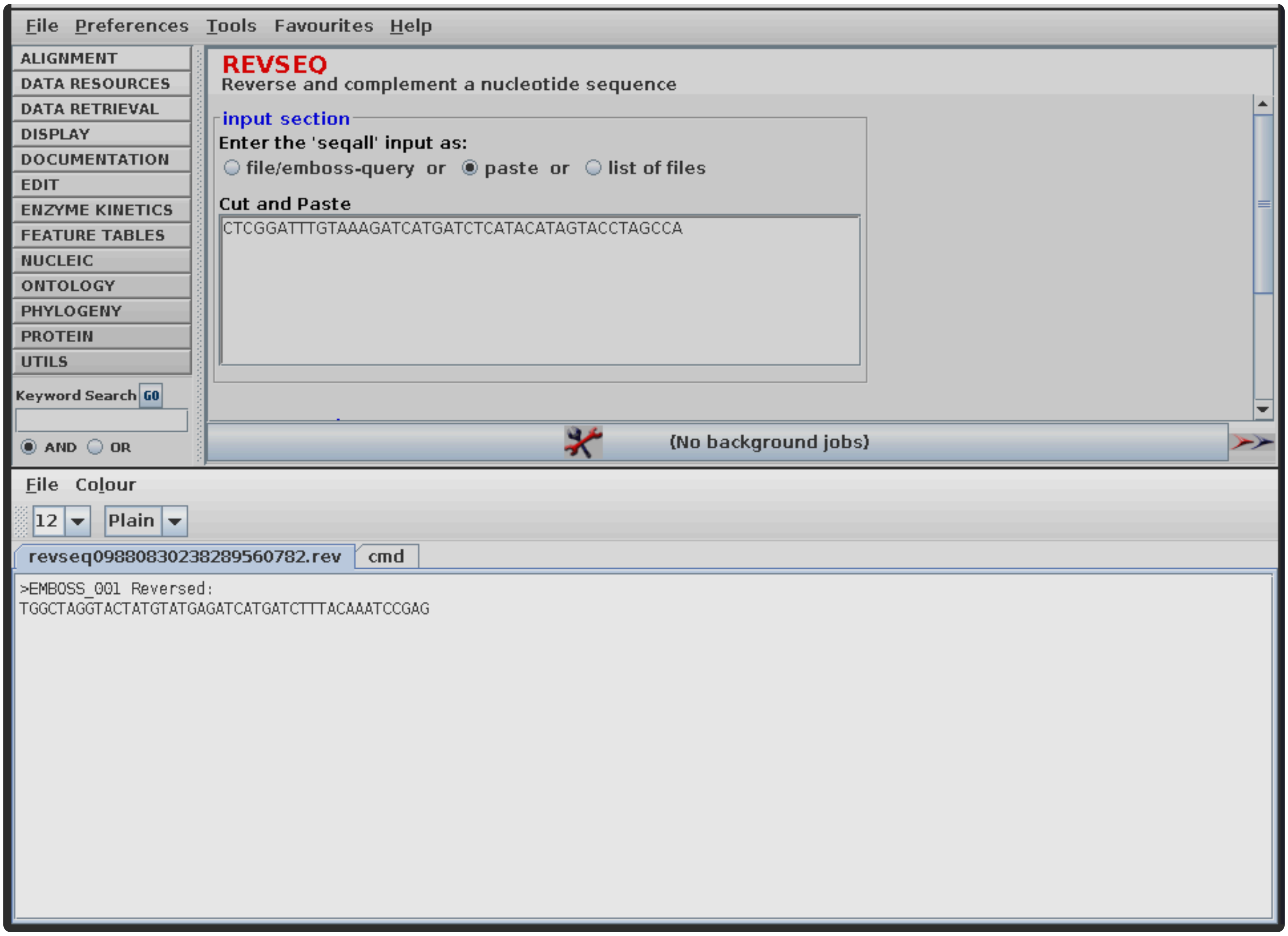
downloaded 👍

Q2 Using EMBOSS, find the complementary strand for the sequence: CTCGGATTTGTAAAGATCATGATCTCATACATAGTACCTAGCCA

using terminal

```
                                                                                          28s 05:13:21
❯ revseq -sequence asis:CTCGGATTTGTAAAGATCATGATCTCATACATAGTACCTAGCCA -outseq stdout -noreverse -complemen
t
Reverse and complement a nucleotide sequence
>asis
GAGCCTAAACATTTCTAGTACTAGAGTATGTATCATGGATCGGT
~ ❯                                                                                      05:22:54 PM
```

using GUI

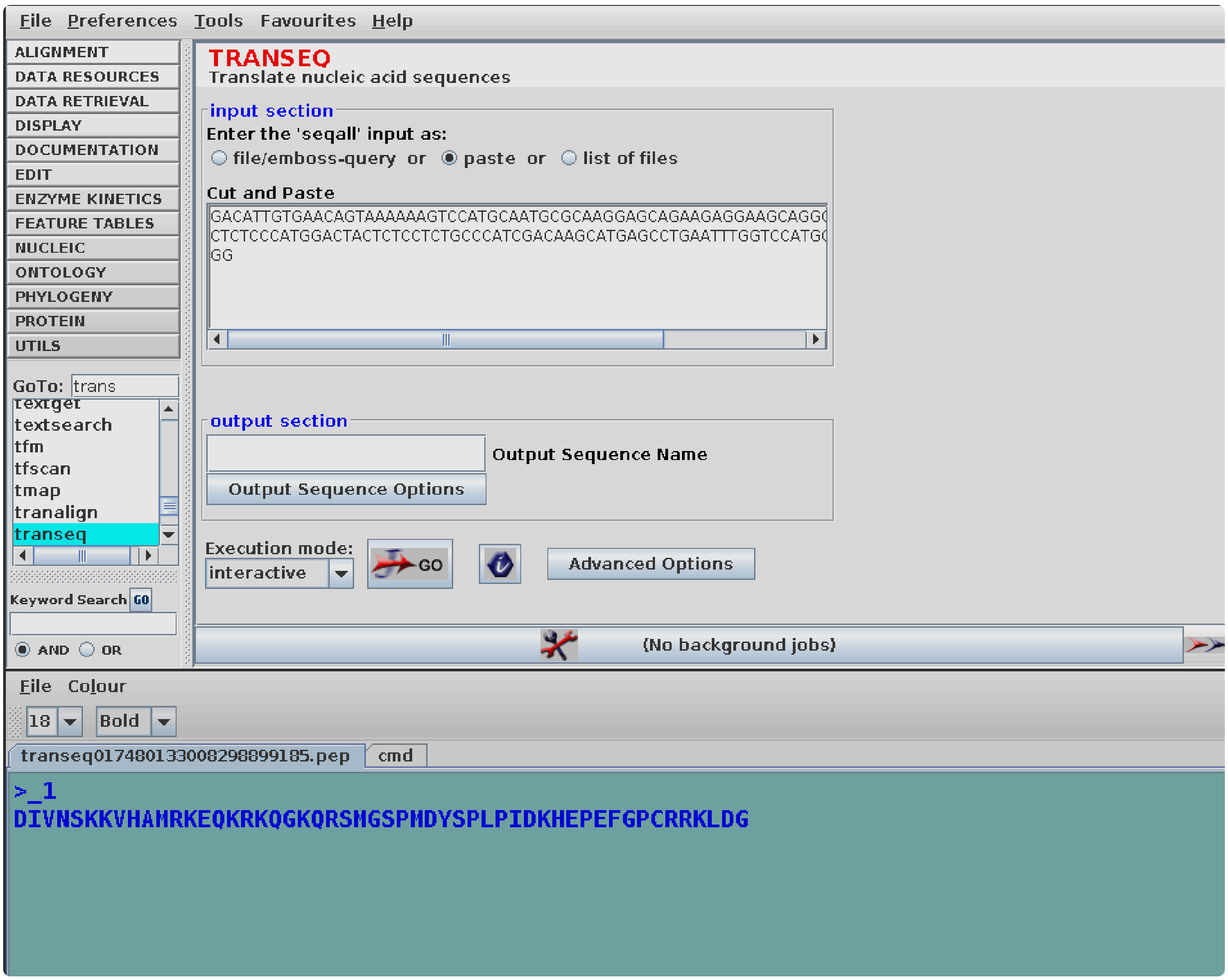


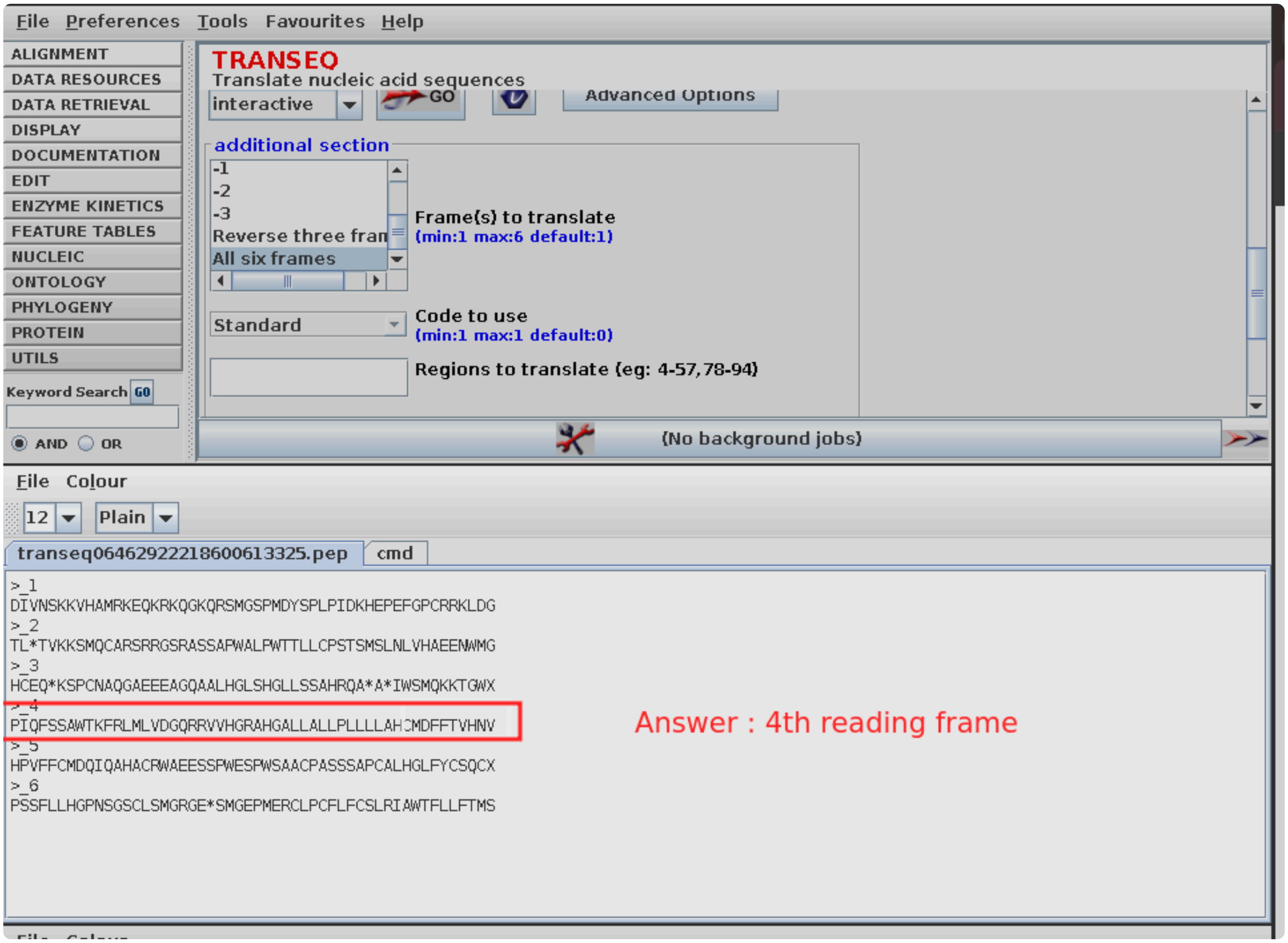
Q3 Write a program to find the complementary strand for the sequence given in Q2.

```
#seq : str = input() #should be string can be typecasted

seq = "CTCGGATTTGTAAAGATCATGATCTCATACATAGTACCTAGCCA"
seq.upper()

complement_map = {'A':'T', 'T':'A', "G":"C", 'C':'G'}

#o(1) comprehension since hashmap
```

```
complementary_strand = "".join([complement_map[base] for
base in seq])

print("given strand is", seq)
print("its complement is", complementary_strand)
```

Q.4





Q.5 Write a program to translate the given DNA sequence (refer Q4) to the protein sequence.

```
def translate_dna(dna_seq):

    codon_table = {
        'ATA':'I', 'ATC':'I', 'ATT':'I', 'ATG':'M',
        'ACA':'T', 'ACC':'T', 'ACG':'T', 'ACT':'T',
        'AAC':'N', 'AAT':'N', 'AAA':'K', 'AAG':'K',
        'AGC':'S', 'AGT':'S', 'AGA':'R', 'AGG':'R',
        'CTA':'L', 'CTC':'L', 'CTG':'L', 'CTT':'L',
        'CCA':'P', 'CCC':'P', 'CCG':'P', 'CCT':'P',
        'CAC':'H', 'CAT':'H', 'CAA':'Q', 'CAG':'Q',
        'CGA':'R', 'CGC':'R', 'CGG':'R', 'CGT':'R',
        'GTA':'V', 'GTC':'V', 'GTG':'V', 'GTT':'V',
        'GCA':'A', 'GCC':'A', 'GCG':'A', 'GCT':'A',
        'GAC':'D', 'GAT':'D', 'GAA':'E', 'GAG':'E',
        'GGA':'G', 'GGC':'G', 'GGG':'G', 'GGT':'G',
```

```
        'TCA':'S', 'TCC':'S', 'TCG':'S', 'TCT':'S',
        'TTC':'F', 'TTT':'F', 'TTA':'L', 'TTG':'L',
        'TAC':'Y', 'TAT':'Y', 'TAA':'*', 'TAG':'*',
        'TGC':'C', 'TGT':'C', 'TGA':'*', 'TGG':'W',
    }

    protein = ""

    for i in range(0, len(dna_seq) - 2, 3):
        codon = dna_seq[i:i+3]
        protein += codon_table.get(codon, "")

    return protein

dna_sequence =
"GACATTGTGAACAGTAAAAAAGTCCATGCAATGCGCAAGGAGCAGAAGAGGAAGCAG
GGCAAGCAGCGCTCCATGGGCTCTCCCATGGACTACTCTCCTCTGCCCATCGACAAGC
ATGAGCCTGAATTTGGTCCATGCAGAAGAAAACTGGATGGG"
#or we can do input() as well
protein_sequence = translate_dna(dna_sequence)

print(f"DNA Sequence: {dna_sequence}")
print(f"Protein Sequence: {protein_sequence}")
```

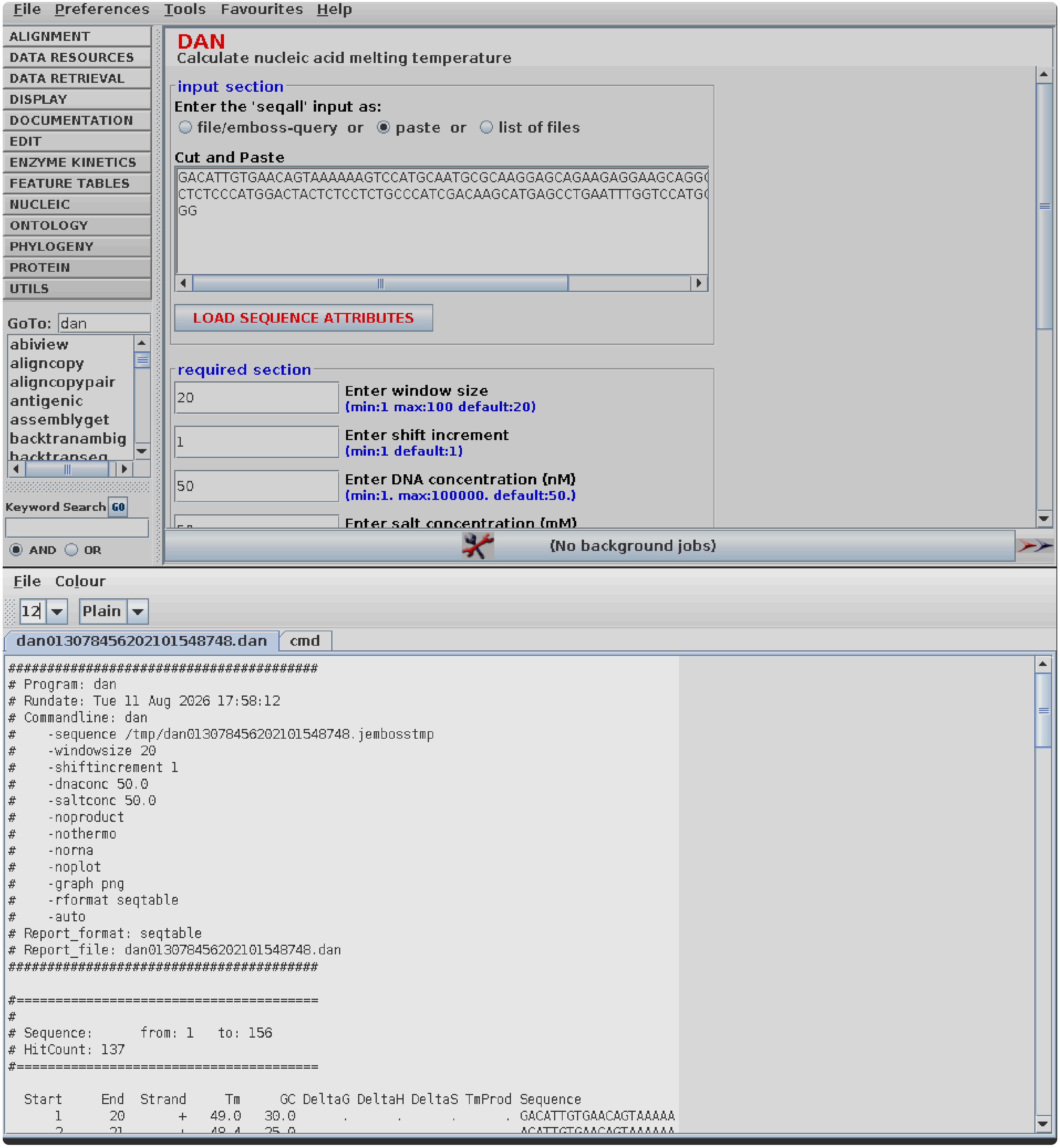
Q.6

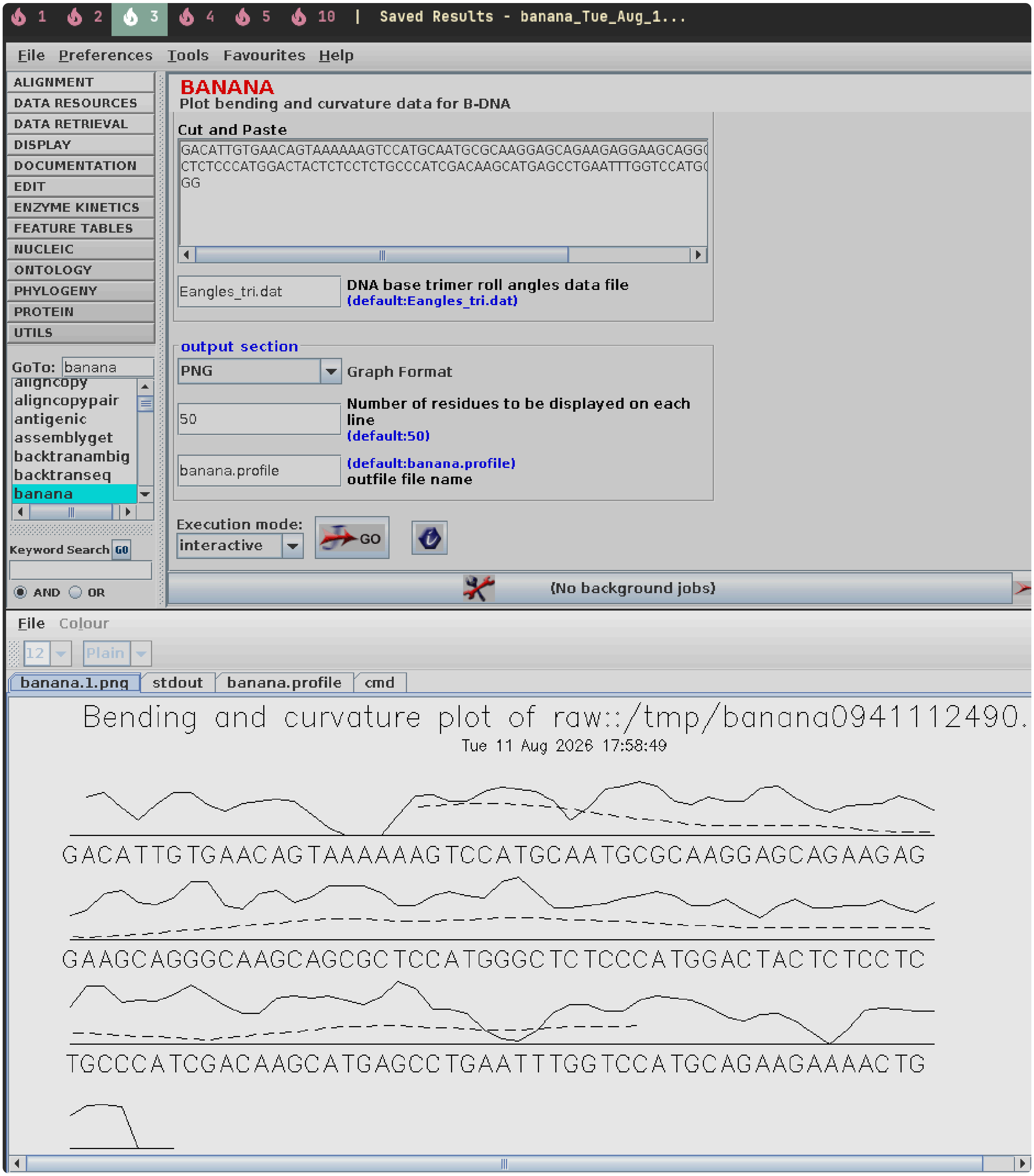
```
# The DNA sequence provided in Question 4
dna_sequence =
"GACATTGTGAACAGTAAAAAAGTCCATGCAATGCGCAAGGAGCAGAAGAGGAAGCAG
GGCAAGCAGCGCTCCATGGGCTCTCCCATGGACTACTCTCCTCTGCCCATCGACAAGC
ATGAGCCTGAATTTGGTCCATGCAGAAGAAAACTGGATGGG"
search_strings = ['AAG', 'GTC', 'GAG', 'ACTA', 'ATAT']

for target in search_strings:

    print(f"Enter the string: {target}")
    positions = []
    index = dna_sequence.find(target)
    while index != -1:
        positions.append(str(index + 1))
```

```
        index = dna_sequence.find(target, index + 1)
    print(f"Total match: {len(positions)}")
    if positions:
        print(f"Position of match: {',
'.join(positions)}")
    else:
        print("Position of match: 0")
```

```
❯ python3 bs.py
Enter the string: AAG
Total match: 7
Position of match: 20, 37, 46, 52, 61, 112, 141
Enter the string: GTC
Total match: 2
Position of match: 22, 131
Enter the string: GAG
Total match: 3
Position of match: 40, 48, 118
Enter the string: ACTA
Total match: 1
Position of match: 89
Enter the string: ATAT
Total match: 0
Position of match: 0
~ ❯
```

Q.7


File Preferences Tools Favourites Help
ALIGNMENT
DATA RESOURCES
DATA RETRIEVAL
DISPLAY
DOCUMENTATION
EDIT
ENZYME KINETICS
FEATURE TABLES
NUCLEIC
ONTOLOGY
PHYLOGENY
PROTEIN
UTILS
GoTo: dan
abiview
aligncopy
aligncopypair
antigenic
assemblyget
backtranambig
Keyword Search GO
AND OR
DAN
Calculate nucleic acid melting temperature
input section
Enter the 'seqall' input as:
file/emboss-query or paste or list of files
Cut and Paste
GACATTGTGAACAGTAAAAAAGTCCATGCAATGCGCAAGGAGCAGAAGAGGAAGCAGG
CTCTCCCATGGACTACTCTCCTCTGCCCATCGACAAGCATGAGCCTGAATTTGGTCCATG
GG
LOAD SEQUENCE ATTRIBUTES
required section
20
Enter window size
(min:1 max:100 default:20)
1
Enter shift increment
(min:1 default:1)
50
Enter DNA concentration (nM)
(min:1. max:100000. default:50.)
(No background jobs)
File Colour
12
Plain
dan013078456202101548748.dan
cmd
##
Program: dan
Rundate: Tue 11 Aug 2026 17:58:12
Commandline: dan
-sequence /tmp/dan013078456202101548748.jembosstmp
-windowsize 20
-shiftincrement 1
-dnaconc 50.0
-saltconc 50.0
-noproduct
-nothermo
-norna
-noplot
-graph png
-rformat seqtable
-auto
Report_format: seqtable
Report_file: dan013078456202101548748.dan
##
#=======================================
#
Sequence: from: 1   to: 156
HitCount: 137
#=======================================
Start End Strand Tm GC DeltaG DeltaH DeltaS TmProd Sequence
1 20 + 49.0 30.0 . . . . GACATTGTGAACAGTAAAAA



Q.8

```
dna_sequence =
"CTCGGATTTGTAAAGATCATGATCTCATACATAGTACCTAGCCA"

energy_matrix = {
```

```
    'AA': -4, 'AT': -7, 'AC': -5, 'AG': -11,
    'TA': -7, 'TT': -2, 'TC': -3, 'TG': -4,
    'CA': -9, 'CT': -5, 'CC': -6, 'CG': -7,
    'GA': -9, 'GT': -6, 'GC': -4, 'GG': -11
}

total_energy = 0
num_pairs = len(dna_sequence) - 1

for i in range(num_pairs):
    dinucleotide = dna_sequence[i:i+2]
    total_energy += energy_matrix[dinucleotide]

average_energy = total_energy / num_pairs

print(f"Sequence: {dna_sequence}")
print(f"Total base stacking energy: {total_energy}")
print(f"Average base stacking energy:
{average_energy:.4f}")
```

Q.9

```
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006$ cd Seq2Feature/
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ ls
data  out_dna  __pycache__  q10_input.tsv  q9_input.tsv  seq2feature_common.py  seq_dna2.py
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ python3 seq_dna2.py --input_file q9_input.ts
v --property PHY --output_dir BS24B006_results
Completed successfully. Output directory: /home/bioinfo/BS24B006/Seq2Feature/BS24B006_results
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ cd BS24B006_results/
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ ls
avg_Physicochemical.csv
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ cat avg_Physicochemical.csv

Input_row,Stacking energy,Enthalpy,Entropy,Flexibility_shift,Flexibility_slide,Free energy,Melting Temper
ature,Mobility to bend towards major groove,Mobility to bend towards minor groove,Probability contacting
nucleosome core,Rise stiffness,Roll stiffness,Shift stiffness,Slide stiffness,Tilt stiffness,Twist stiffn
ess
1,1.62,5.44,14.96,2.277,8.697,0.59,43.202,0.988,0.93,6.08,7.027,17.4,0.803,2.395,25.2,23.3
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ ^C,28.4,18.1
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ ls
avg_Physicochemical.csv
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ cd ..
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ ls
BS24B006_results  data  out_dna  __pycache__  q10_input.tsv  q9_input.tsv  seq2feature_common.py  seq_dna
2.py
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ cat q9_input.tsv
SEQ
ATATATATAT
GCGCGCGCGC
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$ cut -d',' -f1,8 BS24B006_results/avg_Physico
chemical.csv
Input_row,Melting Temperature
1,43.202
2,97.08
(base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature$
```

so the output says the melting temprature of

A is 43.202

B is 97.08

which is expected since B is more GC rich

Q.10

```
at: avg_nu: No such file or directory
base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$ cat avg_Nucleotide.csv
nput_row,GC_content,Pyrimidine_TC_content,Purine_AG_content,Keto_GT_content,Adenine_content,Guanine_content,Cytosine_content,Thymine_content
,41.666667,41.666667,58.333333,33.333333,41.666667,16.666667,25.0,16.666667
base) bioinfo@sudha-ThinkCentre-M82:~/BS24B006/Seq2Feature/BS24B006_results$
```

`AAATGGCCCTAA`

- **GC Content:** `41.67%` (or `41.666667%`)
- **AT Content:** `58.33%` (or `58.333333%`)